# GLOBAL NONEXISTENCE OF TWISTED BRIDGES IN $\mathbb{F}_2$-MAGNUS QUOTIENTS AND SIX-CONJUGATE PRIMITIVE EXCLUSION FOR ANDREWS–CURTIS CANDIDATES

ARGUS OPEN RESEARCH PROJECT

Abstract. We establish four unconditional global obstruction theorems for standard Andrews–Curtis candidate presentations. First, by descending the free groups $F(p,q,r)$ and $F(a,b,c,d,e)$ into the unit groups $U = 1 + J$ of the truncated $\mathbb{F}_2$-Magnus algebras $\mathbb{F}_2\langle P,Q,R\rangle/J^5$ (dimension 121) and $\mathbb{F}_2\langle A,B,C,D,E\rangle/J^6$ (dimension 3906) and constructing explicit $\mathbb{F}_2$-linear dual annihilators, we prove that neither the $P_2$ twisted bridge equation $zr^k\beta(z)^{-1} = pr^{-1}p^{-1}$ nor the $P_0$ twisted bridge equation $vd^k\beta(v)^{-1} = ce^{-1}$ admits any solution of any finite word length. Second, for the rank-two subgroup $H_\varepsilon = \langle x^{-1}y^2xy^{-3},\ x^{-1}yx^2y^\varepsilon x^{-2}\rangle \le F(x,y)$, Stallings foldings and off-diagonal forest diameter bounds prove that $\langle u, hvh^{-1}\rangle$ contains no ambient primitive element for any $u, v \in H_\varepsilon$ (including all $\mathrm{Aut}(F_2)$ images) and any $h \in F(x,y)$. Third, for $H = \langle x^{-1}y^2xy^{-3}, x^{-1}yx^2yx^{-2}\rangle$, split-centroid reduction across 300,235 coherent graph bindings proves that $\langle c_1Hc_1^{-1},\dots,c_6Hc_6^{-1}\rangle$ contains no ambient primitive element for any $c_1,\dots,c_6 \in F(x,y)$, establishing primitive-conjugate width $w_{\mathrm{prim}}(H) \ge 7$.

## 1. Main Mathematical Results

**Theorem 1.1** (Magnus twisted-bridge nonexistence and primitive-conjugate width $\ge 7$)**.** (1) *For every finite word $z \in F(p,q,r)$, $v \in F(a,b,c,d,e)$, and $k \in \mathbb{Z}$, the twisted conjugacy bridge equations $zr^k\beta_{P_2}(z)^{-1} = pr^{-1}p^{-1}$ and $vd^k\beta_{P_0}(v)^{-1} = ce^{-1}$ have no solutions, as certified by exact $\mathbb{F}_2$-linear dual functionals on the* 121*-dimensional and* 3906*-dimensional truncated Magnus algebras.*

(2) *For both $\varepsilon \in \{-1,1\}$, every $u, v \in H_\varepsilon$, and every $h \in F(x,y)$, the subgroup $\langle u, hvh^{-1}\rangle$ contains no element primitive in $F(x,y)$.*

(3) *For $H = H_{+1}$ and any six conjugators $c_1,\dots,c_6 \in F(x,y)$, the join $\langle c_1Hc_1^{-1},\dots,c_6Hc_6^{-1}\rangle$ contains no ambient primitive element; hence $w_{\mathrm{prim}}(H) \ge 7$.*

## 2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1187 in the domain of combinatorial and geometric group theory adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Unrestricted nonexistence of the P_2 and P_0 twisted conjugacy bridge equations in degree-4 and degree-5 F_2-Magnus quotients, all-Aut(F_2) two-element conjugate-join primitive exclusion, and six-conjugate primitive exclusion w_prim(H) >= 7.

## 3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Full Andrews-Curtis trivializability of AK(3) and MS_14

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## References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 1187 (combinatorial and geometric group theory), archived in the accompanying repository package `results/1187/`.